

AI Notes DAY-1(NumPy)

1. What is NumPy?

NumPy (Numerical Python) is a Python library used for numerical computing. It provides fast and efficient multidimensional arrays and mathematical functions, making it useful for AI, Machine Learning, Data Science, and Scientific Computing.

2. Why do we use NumPy?

To perform fast mathematical calculations.

To work with large datasets efficiently.

To create and manipulate arrays.

To perform matrix operations.

It is widely used in AI, Machine Learning, and Data Analysis.

3. Advantages of NumPy

Faster than Python lists.

Uses less memory.

Supports multidimensional arrays.

Provides many built-in mathematical functions.

Easy integration with Pandas, Matplotlib, TensorFlow, and Scikit-learn.

4. Important Functions

`np.array()`

Creates a NumPy array.

`import numpy as np`

```
arr = np.array([1,2,3,4])
```

```
print(arr)
```

```
np.arange()
```

Creates an array with a range of values.

```
arr = np.arange(1,10)
```

```
print(arr)
```

```
np.zeros()
```

Creates an array filled with zeros.

```
arr = np.zeros((2,3))
```

```
print(arr)
```

```
np.ones()
```

Creates an array filled with ones.

```
arr = np.ones((2,2))
```

```
print(arr)
```

```
np.reshape()
```

Changes the shape of an array.

```
arr = np.arange(1,7)
arr = arr.reshape(2,3)
print(arr)
np.shape()
```

Returns the shape (rows and columns) of an array.

```
print(arr.shape)
```

5. Indexing

Indexing is used to access a specific element in an array.

```
import numpy as np
```

```
arr = np.array([10,20,30,40])
```

```
print(arr[2])
```

Output

30

6. Slicing

Slicing is used to access multiple elements from an array.

```
import numpy as np
```

```
arr = np.array([10,20,30,40,50])
```

```
print(arr[1:4])
```

Output

[20 30 40]

7. Examples

Example 1

```
import numpy as np
```

```
arr = np.array([5,10,15,20])
```

```
print(arr)
```

Example 2

```
import numpy as np
```

```
matrix = np.array([[1,2],[3,4]])
```

```
print(matrix)
```

Example 3

```
import numpy as np
```

```
arr = np.arange(1,11)
```

```
print(arr)
```

8. Interview Questions

Q1. What is NumPy?

NumPy is a Python library for numerical computing and array operations.

Q2. Why is NumPy faster than Python lists?

Because NumPy stores data in contiguous memory and is implemented in optimized C code.

Q3. What is the difference between a Python list and a NumPy array?

Python lists can store different data types, while NumPy arrays usually store the same data type and are faster for mathematical operations.

Q4. What is indexing?

Indexing is used to access a single element of an array.

Q5. What is slicing?

Slicing is used to access a range of elements from an array.

9. My Learning

Today, I learned the basics of NumPy, including creating arrays, generating arrays with `arange()`, creating arrays of zeros and ones, reshaping arrays, checking array shapes, indexing, and slicing. I also understood why NumPy is faster than Python lists and why it is widely used in AI, Machine Learning, Data Science, and Scientific Computing. This knowledge will help me build a strong foundation for future AI and ML projects.